

东南大学本科生考试试卷（期中卷）

课程代码: B0402010 课程名称: 计算机组织与结构 (双语) I
学年学期: 2025-2026 学年 秋季学期 考试方式: 闭卷
考试时长: 90 分钟

一卡通号: 学号: 姓名:

请在答题卡上作答, 在试卷上作答无效

一、Fill blanks (Total 13 blanks, 2 points/blank, total 26 points)

- Many computer manufacturers offer a family of computer models, all with the same ① but with differences in ②. Consequently, the different models in the family have different price and performance characteristics.
- When several processors all reside on a single chip, such as IBM zEnterprise EC12 processor unit chip, the term ③ computer is used, and each processing unit (consisting of a control unit, ALU, registers, and perhaps cache) is called a ④.
- The transfer unit between cache and the main memory is a ⑤, which consists of some fixed number of words.
- The processor accesses the ⑥ cache directly, without going through the memory management unit (MMU).
- As one goes down the memory hierarchy, the frequency of access of the memory by the processor will decrease. The basis for the validity of this condition is a principle known as ⑦. There is one addressing mode that exploits this principle, called ⑧.
- There are three common places for storing the return address of the procedure call: (1) ⑨, (2) Start of called procedure, (3) ⑩.
- List four processor's instruction cycle: Fetch ⑪, Execute, and ⑫.
- Assume that the access time for a cache is 2 ns and the access time for a SDRAM used as main memory is 50 ns. In order to achieve an average access time of 3 ns for the 2-level storage system, the cache hit ratio should be ⑬.

二、Translate into Chinese (Total 15 points)

- (5 points) In a class of computers called microcomputers, the relationship between architecture and organization is very close. Changes in technology not only influence organization but also result in the introduction of more powerful and more complex architectures. Generally, there is less of a requirement for generation-to-generation compatibility for these smaller machines.

- (5 points) One of the most interesting, and most analyzed, aspects of computer design is instruction set design. The design of an instruction set is very complex because it affects so many aspects of the computer system. The instruction set defines many of the functions performed by the processor and thus has a significant effect on the implementation of the processor.

- (5 points) Guessing can reduce the time loss from the second reason. A simple rule is the following: When a conditional branch instruction is passed on from the fetch to the execute stage, the fetch stage fetches the next instruction in memory after the branch instruction. Then, if the branch is not taken, no time is lost. If the branch is taken, the fetched instruction must be discarded and a new instruction fetched.

三、Review Questions (Total 17 points)

- (4 points) List the main structural components of a computer and a processor.
- (4 points) What are the differences among direct mapping, associative mapping, and set-associative mapping?
- (4 points) The loop buffer is similar in principle to a cache dedicated to instructions. What are the differences between the loop buffer and the cache?
- (5 points) The branch history table can also be treated as a small cache and it consists of three elements in the table. Please briefly describe the purpose of the branch history table, and list three elements of the table.

四、Problems (Total 42 points)

- (12 points) The main memory address space of a computer is 256 MBytes, addressed in bytes. The instruction cache and data cache are separated. Each cache has 8 lines and the line size of 64 Bytes. The data cache uses direct mapping method. There are two programs A and B with the same function, and they are as follows:

Program A

```
int x[256][256];
int i, j, sum=0;
for (i=0; i<256; i++) {
    for (j=0; j<256; j++) {
        sum+=x[i][j];
    }
}
```

Program B

```
int x[256][256];
int i, j, sum=0;
for (j=0; j<256; j++) {
    for (i=0; i<256; i++) {
        sum+=x[i][j];
    }
}
```

Assume that int-type data is represented using 32-bit two's complement, and that during program compilation, i , j , and sum are all allocated in registers. The array x is stored in row-major order (elements of a 2D array are stored row by row in contiguous memory locations). The starting address of the array is 320 (decimal), which is 140 in hexadecimal. Please answer the following questions, providing reasons or calculation steps.

1) Assume that the total capacity of the data cache is the sum of two parts: the Tag field and the Data field. Please calculate the total capacity of the data cache.

2) What are the cache line numbers corresponding to the array elements $a[0][16]$ and $a[1][1]$? (Cache line numbers start from 0)

3) What are the data access hit ratios for Program A and B, respectively? Which program has a shorter execution time?

2. (6 points)

1) Convert the following formula from infix to reverse Polish:

$$X = (A - B \times C) / (D + E \times Y)$$

2) Write programs to compute the above formula using a stack.

3. (16 points) The word length of a computer is 16 bits, and the main memory is addressed by word (2 Bytes). The instruction format is as follows:

15	10	9	8	7	0
Opcode			M		A

Among them, A is the address field; M specifies an addressing mode.

M=00b: Direct addressing.

M=01b: use index register X1 for indexing.

M=10b: use index register X2 for indexing.

M=11b: Relative addressing.

Assuming (PC)=1233H (before PC update), (X1)=0036H, (X2)=1121H (H represents hexadecimal number).

Please determine the effective address (EA) of the following 4 instructions and complete the table below.

Instructions (hex)	M (binary)	A (hex)	The formula of EA (e.g. EA=R1)	EA (hex)
4420	①	②	③	④
2245	⑤	⑥	⑦	⑧
1323	⑨	⑩	⑪	⑫
3522	⑬	⑭	⑮	⑯

4. (8 points) Consider an instruction sequence of length n that is streaming through the instruction pipeline of 6 stages (FI, DI, CO, FO, EI, WO).

1) Draw a timing diagram to show the pipeline for the case that a branch is taken in the 3th instruction after EI and jumps to instruction 10.

2) Let the probability of encountering a conditional or unconditional branch instruction be $1/6$, and let the probability that execution of a branch instruction causes a jump to a nonconsecutive address be $1/2$. Assume that each such jump requires the pipeline to be cleared, destroying all ongoing instruction processing from the end of EI stage. Calculate the speedup factor for the pipeline compared to execution without the pipeline.